California Housing: A Visualization Story of Learning Geometry

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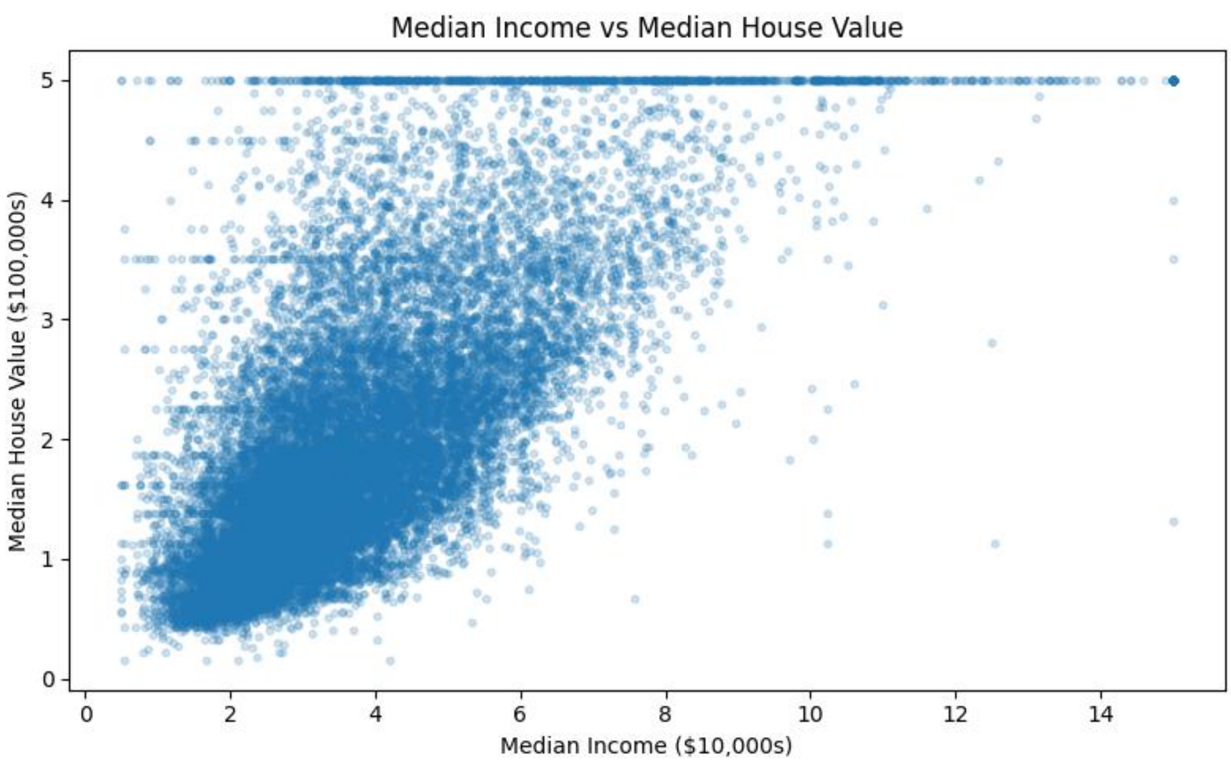
1. Question and Data Familiarization

This study uses the California Housing dataset to examine how data structure changes the behavior of prediction methods. The analysis begins with a simple question: how strongly does median income explain median house value, and what important features of the data become visible before modeling?

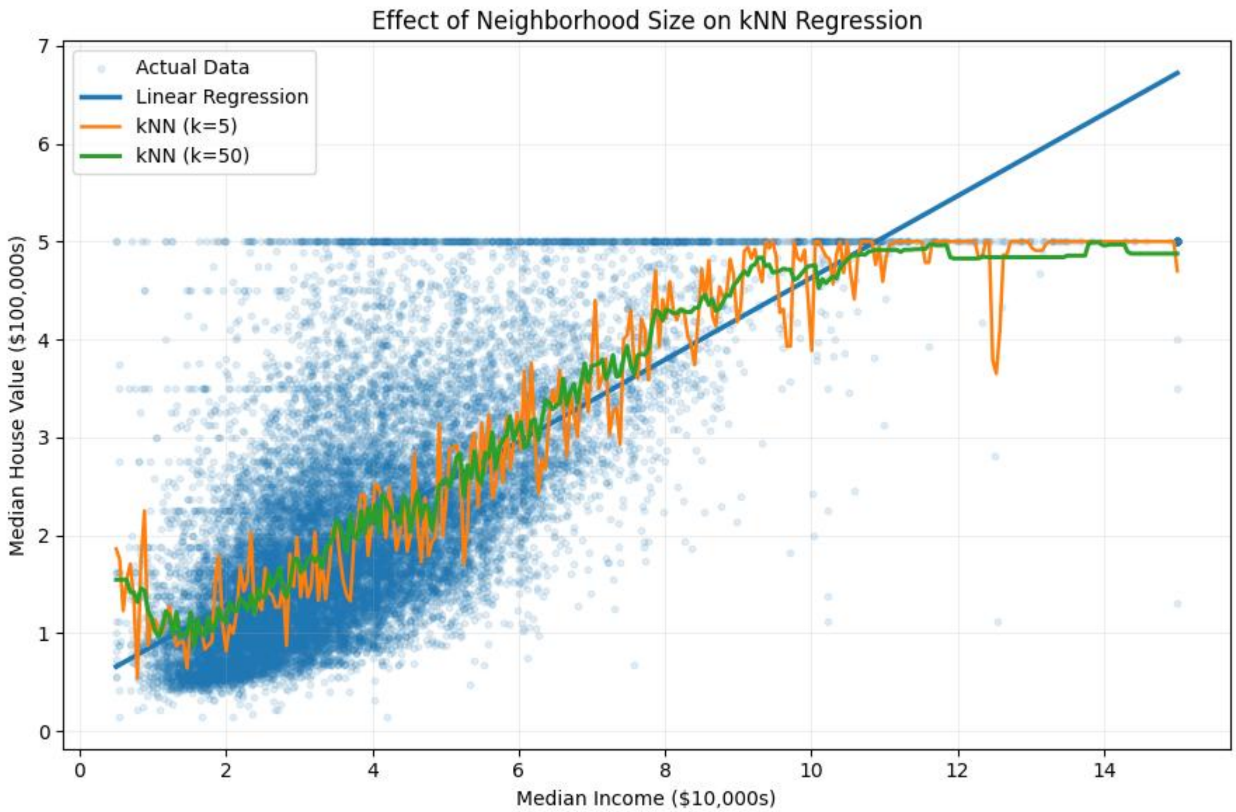
The dataset contains 20,640 districts, eight numerical input features, and median house value as the response. The response is right-skewed and has a clear ceiling near 5.0 (about $500,000). Median income is also right-skewed. The income-value relationship is positive, but the large vertical spread shows that income alone cannot explain housing prices.

Initial insight: the target ceiling is not just a plotting detail; it constrains the response and influences what a fitted model can learn at the high-value end.



2. Global vs Local Learning

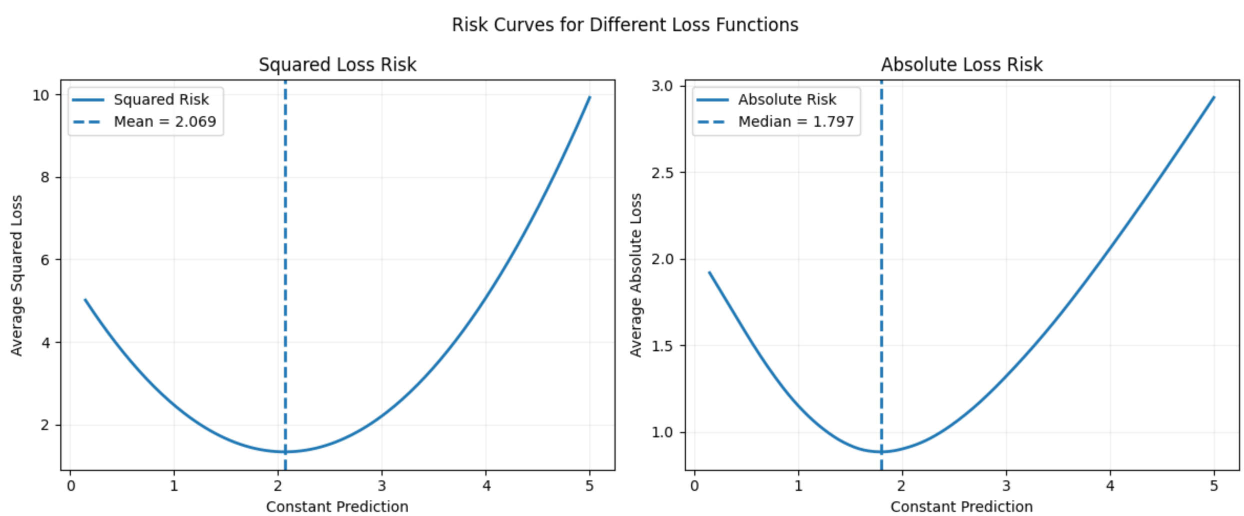
Using median income as the input, a linear regression model was compared with k-nearest-neighbor (kNN) regression. Linear regression imposes one global straight-line relationship. kNN instead predicts from nearby observations, so its shape depends strongly on the neighborhood size.



Interpretation: small neighborhoods provide local flexibility but are sensitive to individual observations. Increasing k sacrifices some local detail for stability. The comparison makes the bias-locality trade-off visible rather than treating model choice as only an error metric.

3. The Loss Function Changes the Best Prediction

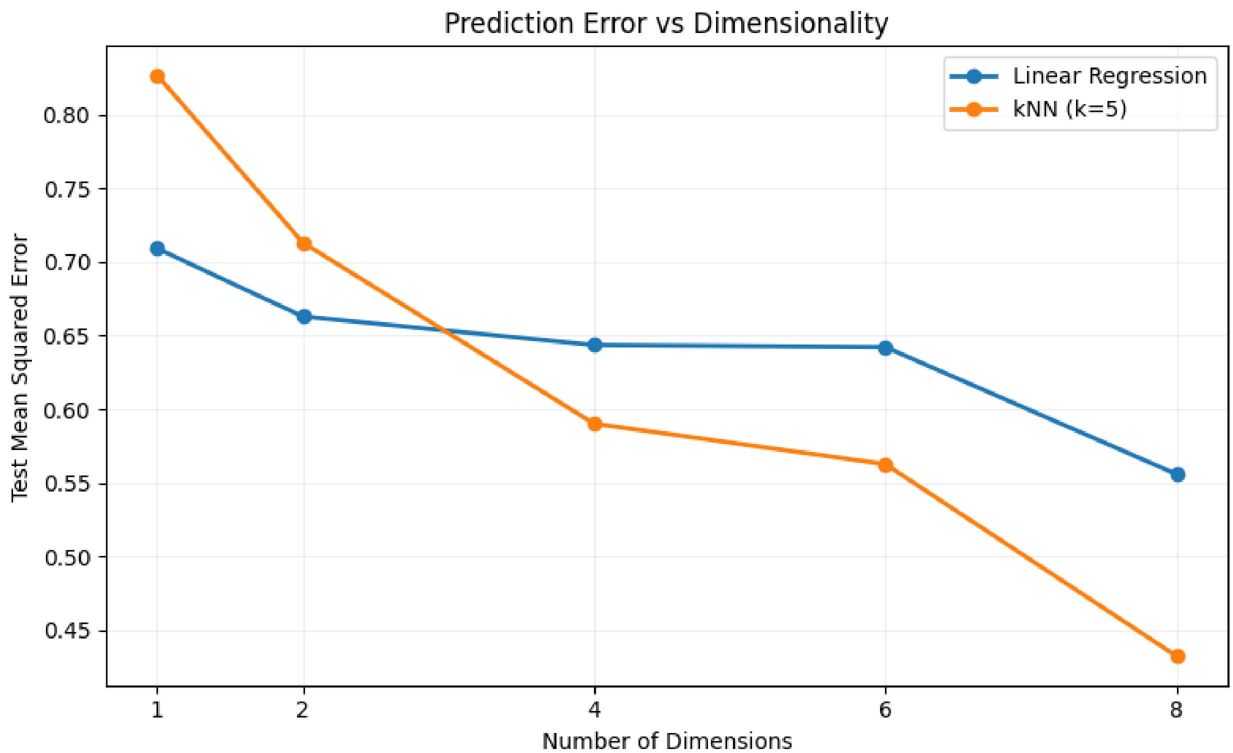
A constant predictor was evaluated under two loss functions. The sample mean house value was 2.069, while the median was 1.797. For each candidate constant, empirical squared-loss risk and absolute-loss risk were calculated.



Interpretation: the 'best' prediction is defined by the loss function. Squared loss penalizes large errors more strongly and therefore targets the mean; absolute loss is less sensitive to extreme values and targets the median. Because the response is skewed, the two optima are visibly different.

4. Dimensionality and Local Geometry

The effective input dimension was increased from 1 to 8 features. After standardization, the average nearest-neighbor distance increased from approximately 0.0002 in one dimension to 0.3435 in eight dimensions. This is the geometric effect behind the curse of dimensionality: points become less locally concentrated as dimension grows.



Important nuance: higher dimension makes nearest-neighbor geometry less local, but prediction error does not have to increase. Here, the added housing variables contain useful information, and their predictive benefit is large enough to offset the geometric disadvantage for kNN.

5. Optional Extension: Structured Linear Models

OLS was compared with Ridge and Lasso using standardized inputs. Their test MSE values were approximately 0.5559, 0.5559, and 0.5483 respectively. Ridge remained very close to OLS, while Lasso shrank coefficients more strongly and reduced the Population coefficient to approximately zero. Five-fold cross-validation favored weak regularization (best Ridge alpha about 0.1; best Lasso alpha about 0.00072). Strong Lasso regularization sharply increased error, indicating underfitting.

Model	Test MSE	Main observation
OLS	0.5559	Unregularized global linear baseline
Ridge	0.5559	Very similar coefficients and error to OLS
Lasso	0.5483	Stronger shrinkage; Population driven to ~0

6. Conclusion

The analysis shows that model behavior is inseparable from data geometry and the objective used to measure error. Median income provides a strong global signal, but local methods reveal structure that a single straight line cannot capture. Changing the loss function changes the optimal constant prediction; increasing dimension changes what nearby means; and regularization changes how strongly a linear model relies on individual features.

The central result is that no modeling principle should be interpreted in isolation. Higher dimensionality makes local neighborhoods geometrically weaker, yet informative features can still improve kNN accuracy. Likewise, regularization can improve stability and sparsity, but excessive regularization can underfit. The visualizations therefore connect prediction performance to the structure of the data rather than treating the models as black boxes.